

Proposed Solution

Project Design Phase · AI/ML and GenAI Track

FitBuddy is a web application. The user enters basic details; the server validates them, asks Gemini for a personalised 7-day plan and a nutrition tip, stores the result, and shows it. The user can then give feedback and receive an updated plan. An admin page lists stored users.

Feature	How it works
Plan generation	The prompt includes name, age, weight, goal and intensity and asks for warm-up, main workout with sets/reps or duration, rest and cooldown, in plain text labelled "Day N - Focus"
Feedback update	Sends the latest plan (updated, or original if none) and the feedback between markers; asks for a complete revised plan
Nutrition tip	A separate small prompt returning 2–4 sentences; if it fails, the plan is still saved
Persistence	A user is created, or an existing user ID is reset with a fresh plan
Resilience	Retries with backoff on 429/5xx errors; friendly error page if Gemini is unavailable
Safety	Prompts avoid extreme exercise, starvation, dehydration and supplements, and encourage gradual progression

Unique value

- Personalised in seconds, at no trainer cost.
- Conversational refinement: the user changes the plan by describing what they want.
- Safety-first prompting for a general audience.